ANY MORPHISM FROM THE SPECTRUM OF AN ARTINIAN RING TO A QUASI-SEPARATED ALGEBRAIC SPACE IS AFFINE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof combines topology (discreteness of Spec of an Artinian ring) with the etale-local criterion for affineness; the finiteness of the fibre is what makes the scheme lemma applicable.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Affineness of a morphism of algebraic spaces is local on the target. Lemma lemma-affine-local, used at proof lines 4154-4156. Reduces the claim to affineness of the fibre products Spec(A) x_X U.
- Quasi-separatedness makes the fibre product over an affine etale U quasi-compact. used at proof lines 4156-4157. Gives quasi-compactness of Spec(A) x_X U.
- Etale morphisms have finite discrete fibres and the spectrum of an Artinian ring is discrete. used at proof lines 4158-4161. Shows the underlying space of the fibre product is finite and discrete.
- A scheme with finite discrete underlying topological space is affine. Schemes, Lemma schemes-lemma-scheme-finite-discrete-affine, used at proof lines 4162-4163. Concludes affineness.

Source reference. Original lemma 03WG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5111–5111. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02O0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3499–3499. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be a quasi-separated algebraic space over S. Let A be an Artinian ring. Any morphism* $\mathrm{Spec}(A) \to X$ *is affine.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $U \to X$ be an étale morphism with U affine. To prove the lemma we have to show that $\mathrm{Spec}(A) \times_X U$ is affine, see Lemma 03WG. Since X is quasi-separated the scheme $\mathrm{Spec}(A) \times_X U$ is quasi-compact. Moreover, the projection morphism $\mathrm{Spec}(A) \times_X U \to \mathrm{Spec}(A)$ is étale. Hence this morphism has finite discrete fibers and moreover the topology on $\mathrm{Spec}(A)$ is discrete. Thus $\mathrm{Spec}(A) \times_X U$ is a scheme whose underlying topological space is a finite discrete set. We are done by Schemes, Lemma 02O0. $\square$